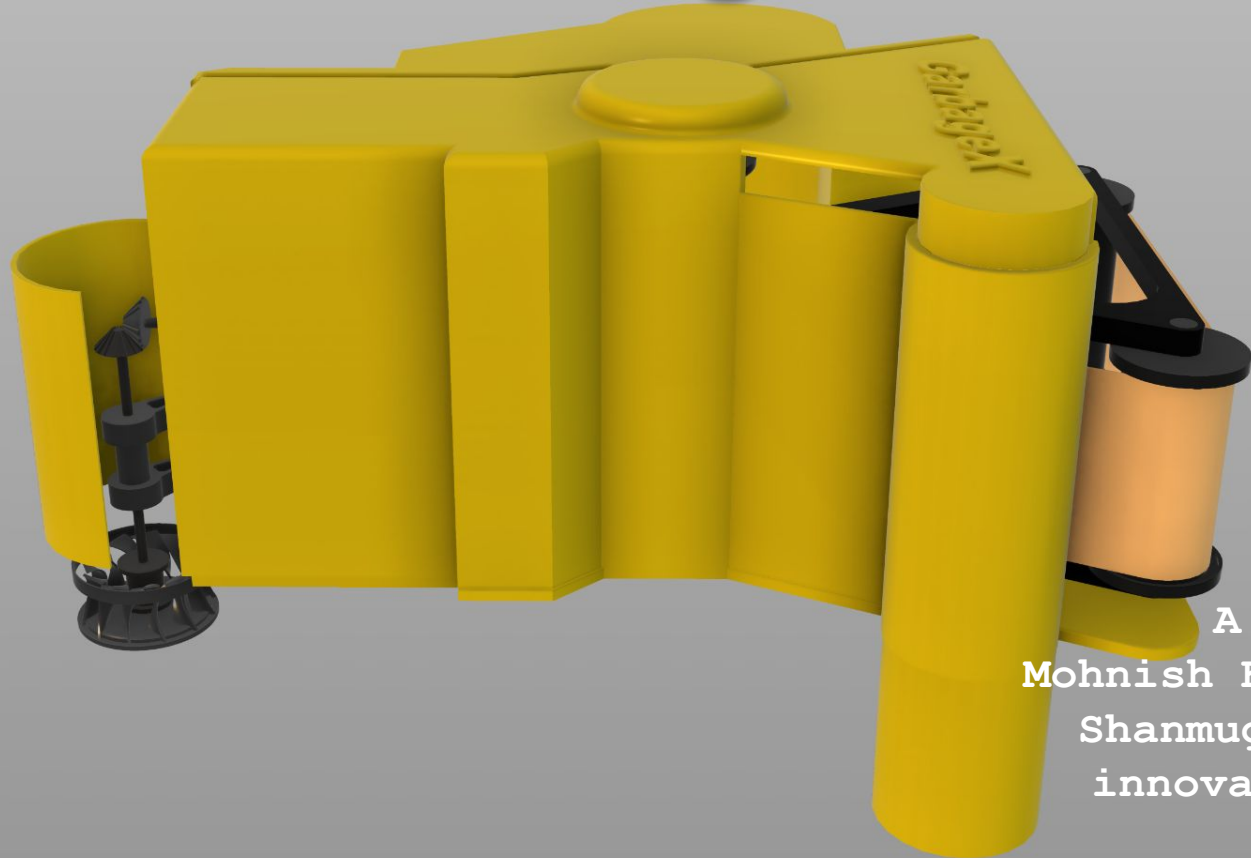


GarbageX



A

Mohnish Kumar.P
Shanmugam.S
innovation

Introduction

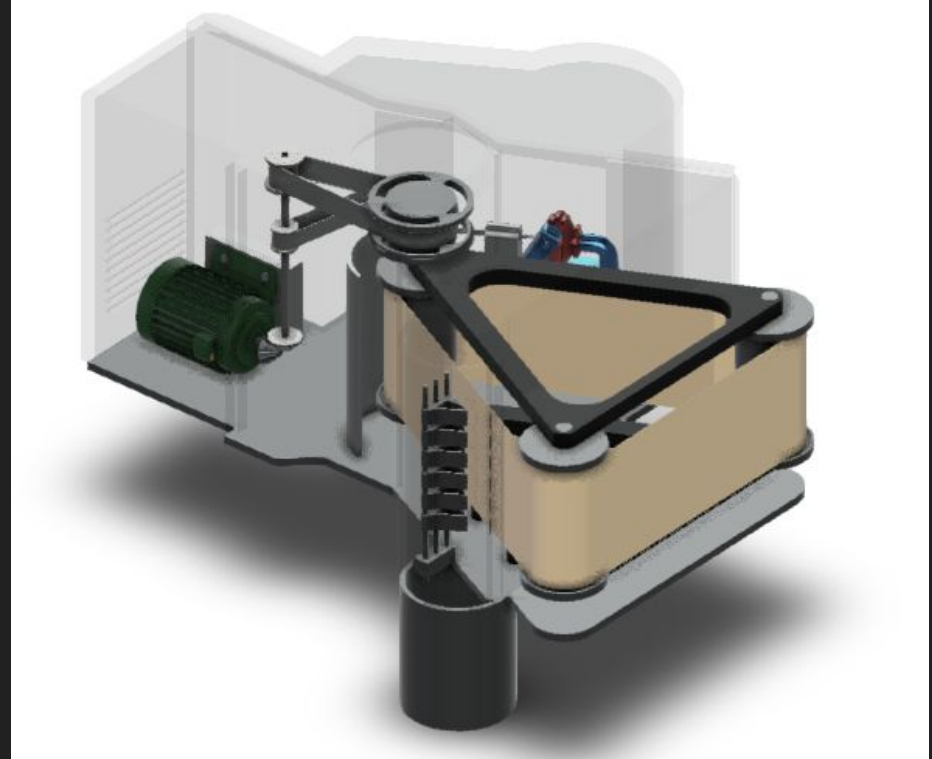
1. The main of the device is to remove the solid waste materials that is floating in the water.
2. The waste material in the surface of the water cause pollution to the wildlife and the environment.
3. It uses the properties of adhesives to remove the waste materials.
4. The removed waste materials is then collected and disposed in a proper way.

Design

The primary part of the device is a belt with three rollers which is used to rotate the belt.

The belt is coated with a waterproof adhesive which stick to the waster materials in the water surface.

The waster materials can be scraped off the belt.



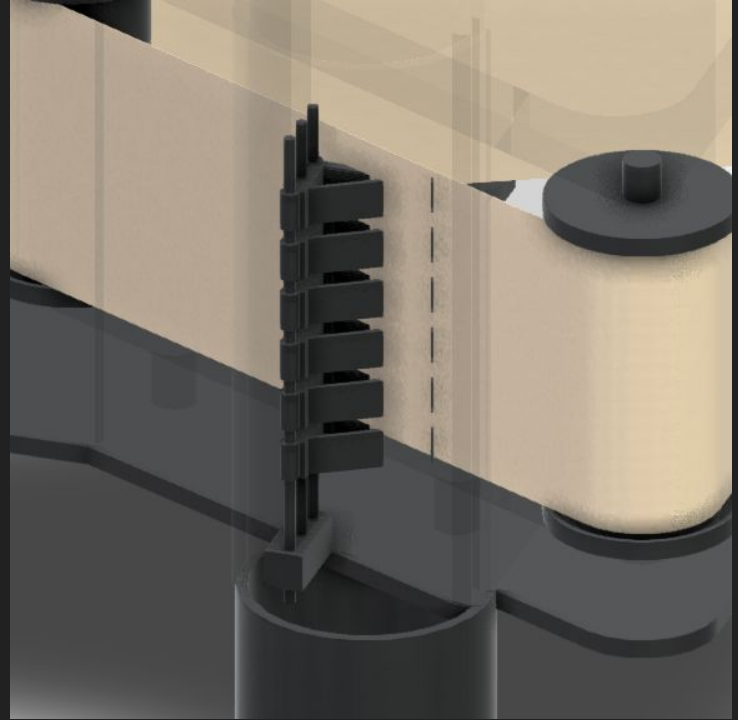
Special components

Scraper :

The scraper is a part which scrapes the waste materials off the belt.

It has wedge shape which is important for the working of the part.

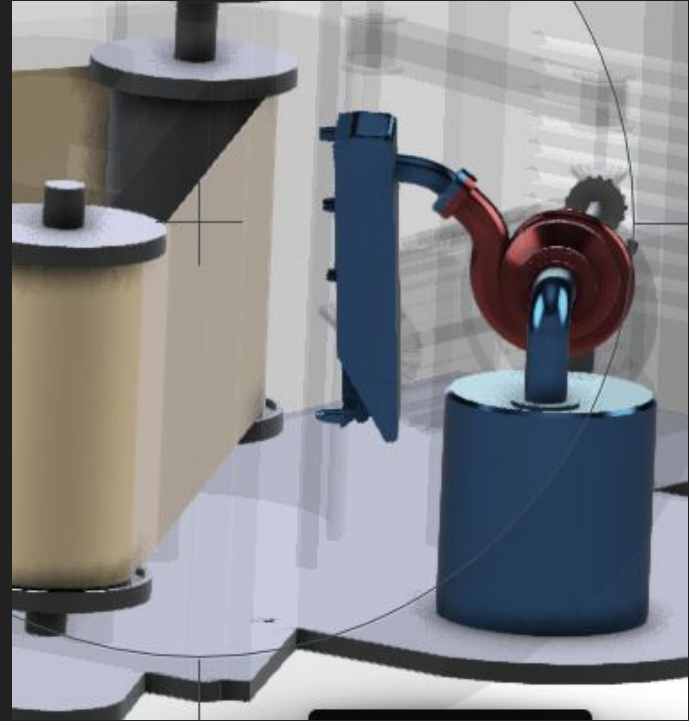
The waste material that is scraped will be collected in a container below.



Adhesive sprayer :

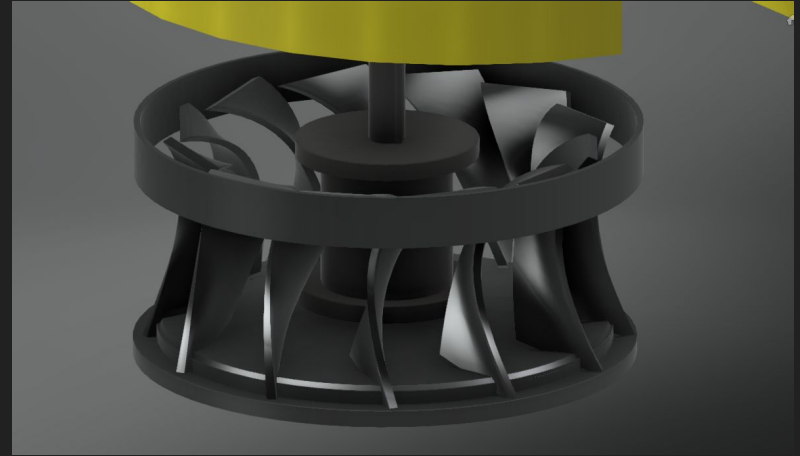
The adhesive in the belt maybe removed or damaged in the scraping process

So a sprayer is used. It consists of a pump , a reservoir and a nozzle. It covers the entire surface of the belt.



Energy recovery system

The bottom of the device houses a turbine which is connected to a generator, when the device is moving a water current is induced and the turbine spins to produce electricity, this electricity is harvested and stored in an onboard battery. The excess energy is then used to run the electric motor for the functioning of the device.



Advantages over conventional methods

1. Compact and less laborious
2. While using conventional methods such as nets, it is not effective against a wide range of size of solid wastes in the water bodies.
3. Larger and smaller objects can be removed since the adhesive property will work on smaller waste material and larger objects won't do damage to the belt and since the larger object has more surface area, it will stick to the belt and can be removed.
4. Faster in action

Conclusion

The solid waste in the water bodies proves to be damaging for the environment. The adhesive solid waste remover is a design that is primarily focused on effectiveness and longevity. It can be deployed in a variety of situations and has wide operating conditions. Using this device we can clean the water bodies which are infested with solid wastes.